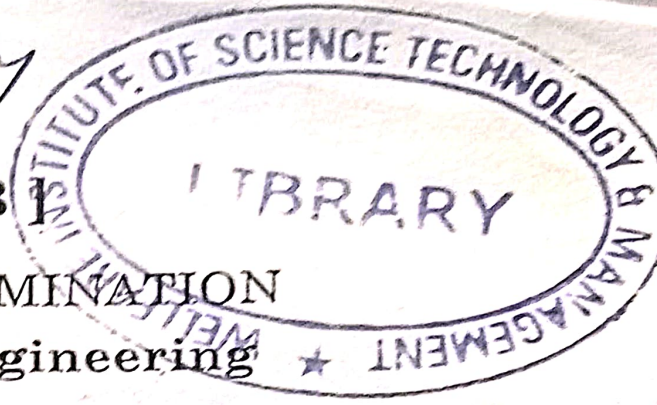


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[07 BENG - 3143]



III/IV B.Tech. DEGREE EXAMINATION

Computer Science and Engineering

First Semester

**DATA COMMUNICATIONS
AND COMPUTER NETWORKS**

(Effective from the admitted batch of 2022 - 2023)

Time : 3 Hours

Max. Marks : 70

Question No. 1 is compulsory.

Answer any FOUR questions from the remaining.

All questions carry equal marks.

Answer all parts of any question at one place.

1. Answer the following in brief:

- (a) Give the formula for Nyquist bit rate.
- (b) Define Protocol.
- (c) What is half - duplex communication?
- (d) What is class E in Internet Protocol?
- (e) Define scrambling.
- (f) Give the components of communication model.
- (g) What is the responsibility of Network Layer?

(P.T.O)

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2. (a) Give the differences between OSI model and TCP/IP Protocol suite.
(b) Explain about Attenuation, Distortion and Noise.
3. (a) Describe the process of synchronous transmission and its advantages over asynchronous transmission.
(b) Explain High - level Data Link Control Protocol (HDLC) in detail.
4. (a) Define Line coding? Draw Unipolar, Polar and Bipolar for taking the sequence 01 00 1110.
(b) Explain about parity check.
5. (a) What is Error detection? Explain CRC with an example.
(b) Write short notes on Go Back N.
6. (a) What is wireless LAN? Explain about IEEE 802.11 with a neat sketch.
(b) Give the frame structure of Wi-Fi.
7. (a) Define congestion. How you can regulate the packet flow with Leaky Bucket Algorithm? Explain.
(b) What are the types of IP addresses?
8. (a) Give the differences between TCP and UDP.
(b) Explain about TCP Header format in detail.

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III/IV B.Tech. DEGREE EXAMINATION.

First Semester

Computer Science and Engineering

DATA COMMUNICATION AND COMPUTER
NETWORKS

(Effective from the admitted batch of 2022-2023)

Time : Three hours

Maximum : 70 marks

First Question is compulsory.

Answer any FOUR from remaining questions.

All questions carry equal marks.

SECTION A — (14 marks)

1. (a) Define Computer Network.
- (b) What is Composite signal?
- (c) What do mean by error control?
- (d) What is packet fragmentation?
- (e) Define Jitter.
- (f) What are tile different transmission media?
- (g) What is FDM?

SECTION B — (56 marks)

2. (a) Write short notes on circuit switching, packet switching and message switching.

(b) Define switching and briefly explain the Datagram Approach in packet switching Method with neat diagram.

3. (a) Explain the working of electronic mail. How SMTP used in email applications?

(b) List and discuss the types of DNS records.

4. (a) Discuss stop and wait protocol.

(b) Discuss about Selective repeat ARQ.

5. (a) Explain distance vector routing in detail.

(b) Discuss about Inter-network Routing.

6. (a) Explain about the Guided transmission Medias in computer networks.

(b) Brief explain about the public switched telephone networks.

7. (a) Write briefly about World Wide Web.

(b) Describe about High-speed LANs.

8. (a) Data Communication is a process of exchanging data or information explains with example.

(b) Define 802.11 Architecture and frame structure with Example.

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III/IV B.Tech. DEGREE EXAMINATION.

First Semester

Computer Science and Engineering

DATA COMMUNICATIONS AND COMPUTER
NETWORKS

(Common with Information Technology)

(Effective from the admitted batch of 2021–2022)

Time : Three hours

Maximum : 70 marks

Answer question No. 1 Compulsory.

Answer any other FOUR Questions from remaining.

All questions carry equal marks.

1. (a) Differentiate simplex and duplex communication systems.
- (b) How data is represented in case of ASCII coding system?
- (c) Write about Digital Signals.
- (d) Differentiate Analog and digital signals.
- (e) Define Piggybacking.
- (f) Write features of UDP.
- (g) What is SMTP?

2. (a) What are the applications and advantages of data communication networks?
- (b) Explain the classification of data communication networks.
3. (a) Write about Block Coding and explain how the errors are detected and corrected using Block coding?
- (b) Explain about HDLC Configurations, Transfer Modes and different types of frames.
4. (a) Find the CRC for the data polynomial $x^9 + x^7 + x^5 + x^2 + 1$, with the generator polynomial $x^3 + x + 1$.
- (b) Differentiate between Packet Switching and Circuit Switching. What kind of networks you will advocate for the PSTN and Internet respectively?
- (a) Suppose the spectrum of a channel is between 5MHz and 6MHz. If the signal to noise ratio is found to be 24dB. Find the maximum number of signal levels in the transmitted signal?
- (b) Draw the block diagram of a general communication model and explain the function of each block?

6. Explain distance vector routing. What are the problems associated with Distance vector routing?
 7. What is Domain Name system? Discuss the three main divisions of the domain name space.
 8. (a) Assume a Go-Back N protocol is used with a window size of 4 and that the ACK for packet 2 gets lost. Show the events until packet 2 is acknowledged at the sender side?
(b) What are the various fields of TCP Header format? Clearly Explain the meaning of each field.
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